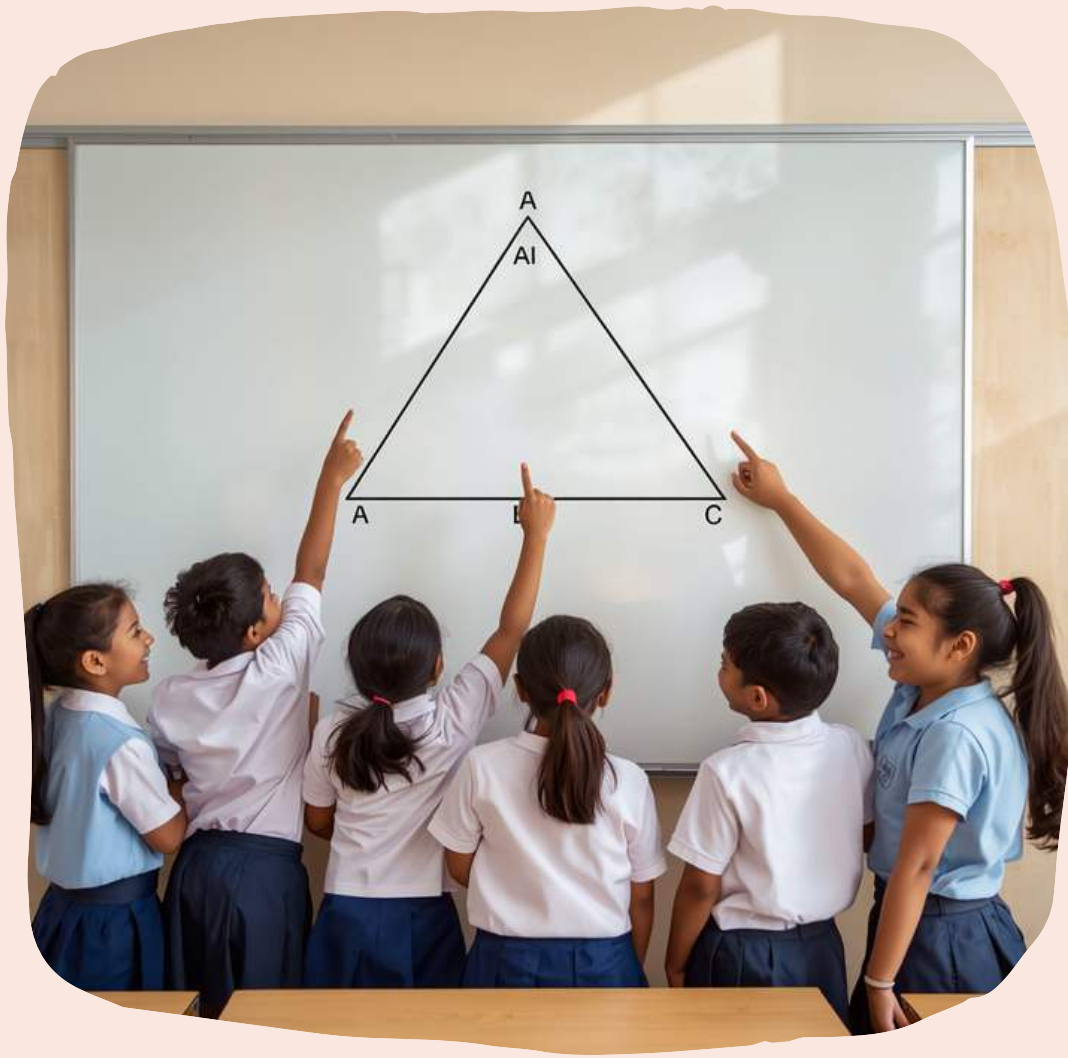


TRIGONOMETRY — SINE RULE & COSINE RULE

Learning with Fun & Curiosity!





What is Trigonometry?

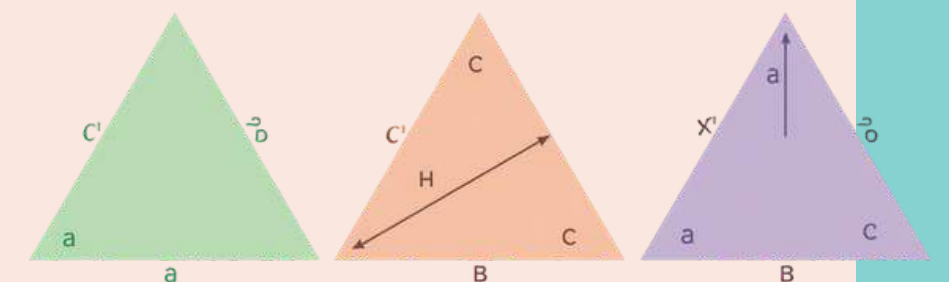
Trigonometry helps us find unknown sides or angles in triangles. It's like being a detective for shapes! When we know some parts of a triangle, trigonometry helps us discover the missing pieces.

Let's Explore!

Types of Triangles & Angles

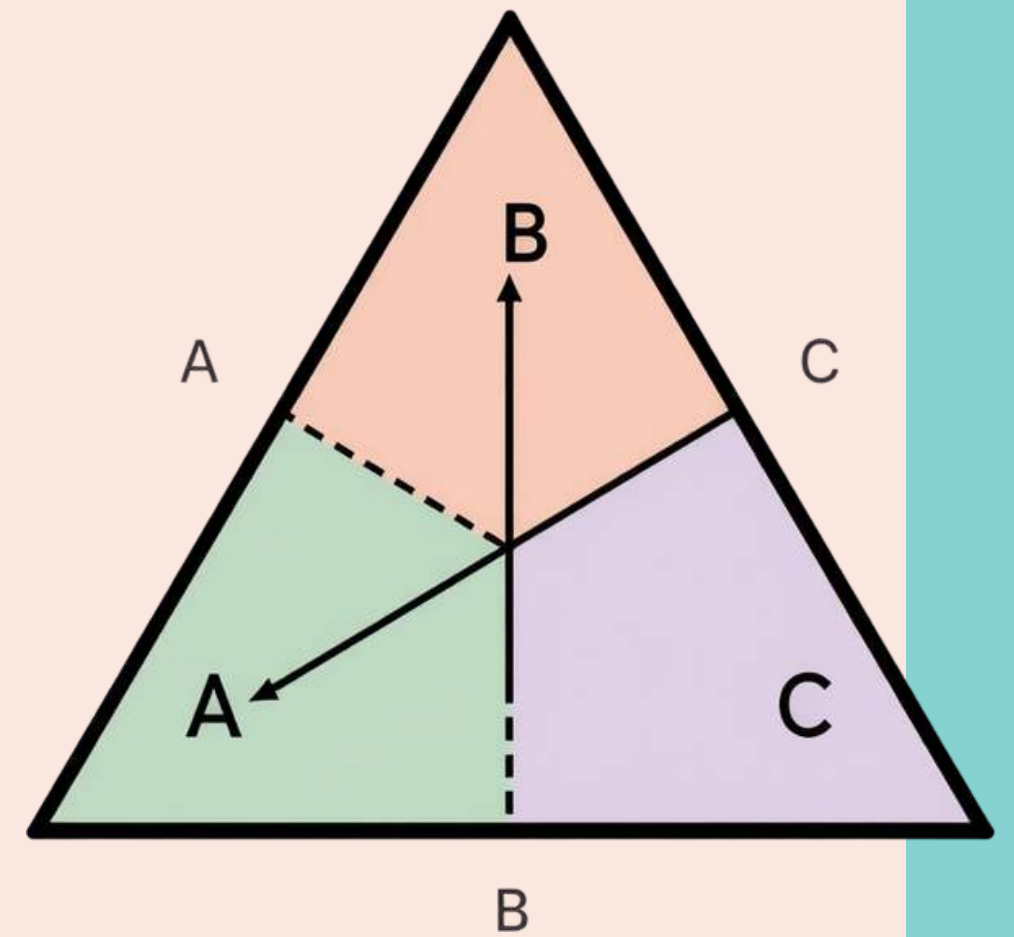
Triangles have 3 sides (a, b, c) and 3 angles (A, B, C)

TRIANGLES



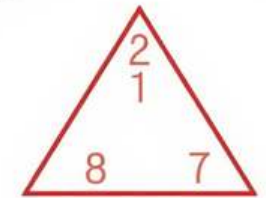
Meet the Sine Rule

$$a/\sin A = b/\sin B = c/\sin C$$



Sine Rule Example

Find side b if $a = 7$ cm,
Angle $A = 50^\circ$, Angle $B = 60^\circ$



When to Use the Sine Rule?

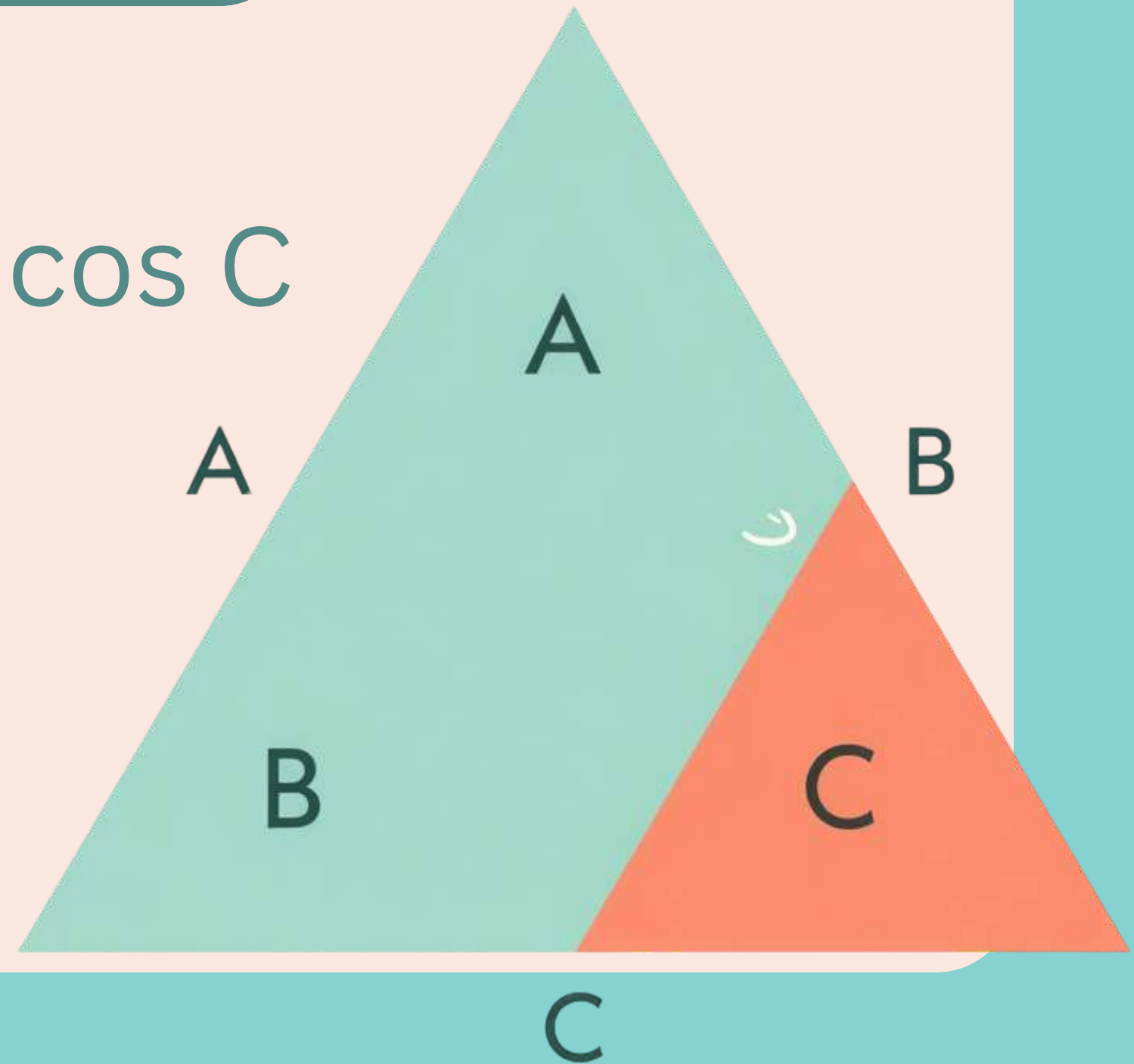
Two angles and one side
known (AAS or ASA)
Two sides and an angle
opposite one of them
(SSA)

Remember: $a/\sin A = b/\sin B = c/\sin C$



Meet the Cosine Rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

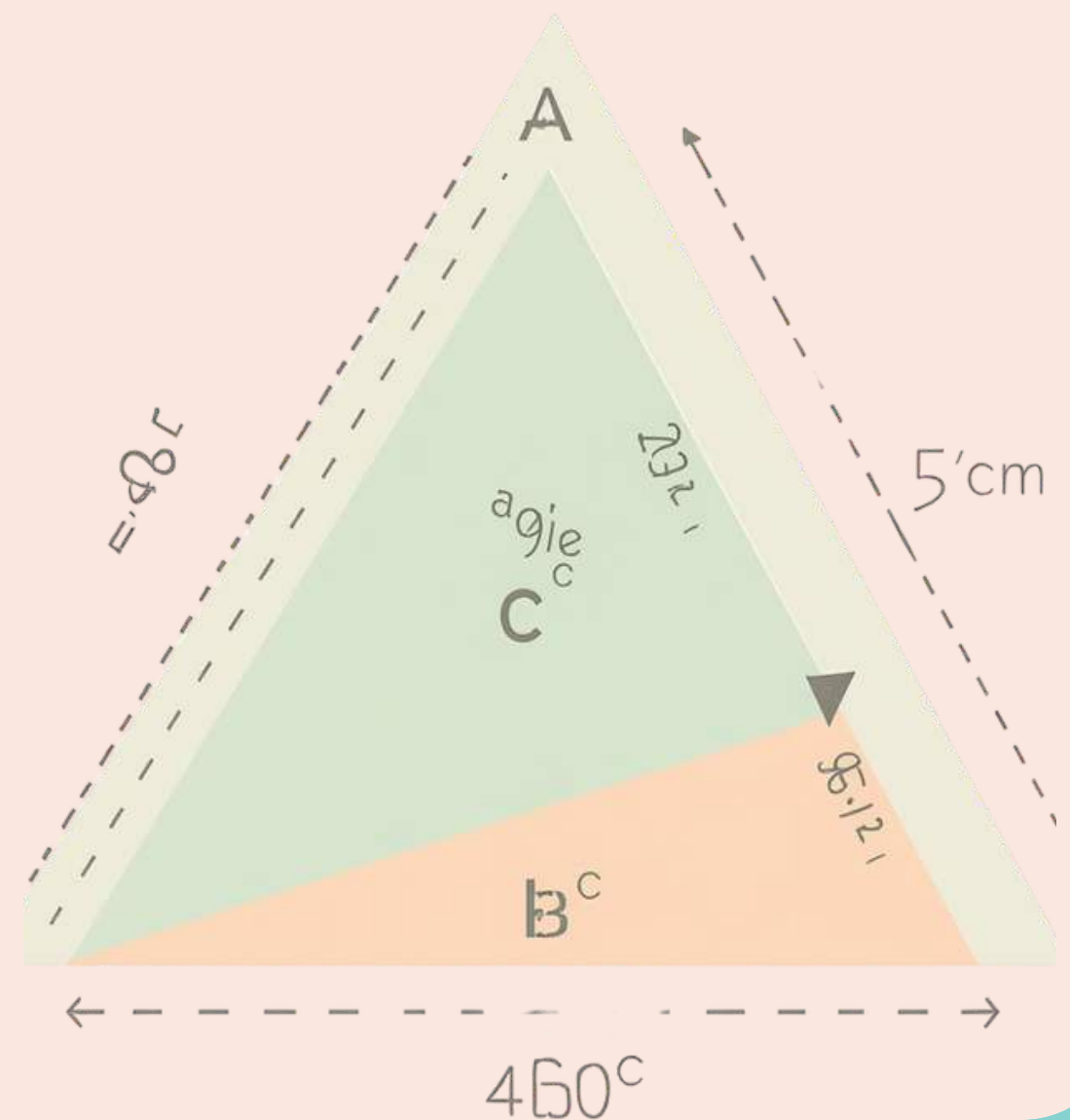


Cosine Rule Example

$$c^2 = 5^2 + 6^2 - 2(5)(6)\cos 60^\circ$$

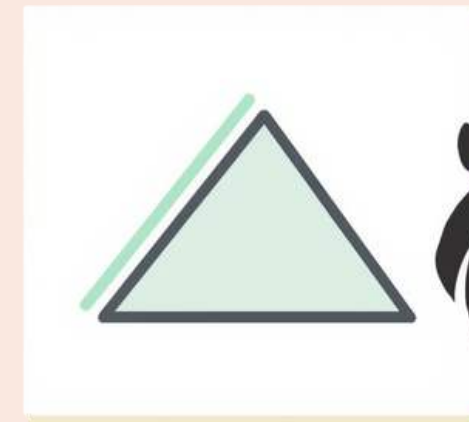
$$c^2 = 25 + 36 - 30 = 31$$

$$c = \sqrt{31} \approx 5.57 \text{ cm}$$



When to Use the Cosine Rule?

SAS: Two sides and the included angle are known
SSS: All three sides are known to find an angle
Use cosine rule when sine rule cannot help!



Sine vs Cosine Rule Comparison

Sine Rule

- Used for AAS, ASA, SSA
- Uses ratios of sides to sines
- Formula: $a/\sin A = b/\sin B = c/\sin C$

Cosine Rule

- Used for SAS, SSS
 - Uses squares of sides & cosine
 - Formula: $c^2 = a^2 + b^2 - 2ab \cos C$
- Choose the right rule for your triangle!



Practice Question 1

Try These Sine Rule Problems!

1. Find side b if $a = 8 \text{ cm}$, $\angle A = 45^\circ$, $\angle B = 70^\circ$
2. Find $\angle B$ if $a = 10 \text{ cm}$, $b = 12 \text{ cm}$, $\angle A = 40^\circ$
3. Find side c if $\angle A = 55^\circ$, $\angle C = 65^\circ$, $a = 9 \text{ cm}$



Practice Question 2

1. Find side c if $a = 8$ cm, $b = 5$ cm, $\angle C = 45^\circ$
2. Find $\angle A$ if $a = 7$ cm, $b = 9$ cm, $c = 6$ cm
3. Find side b if $a = 10$ cm, $c = 12$ cm, $\angle B = 70^\circ$





Trigonometry in
Action

Real-life Applications of Trigonometry

Architecture: Engineers use sine and cosine rules to calculate roof angles and building heights.

Navigation: Sailors and pilots use these rules to find distances and directions.

Sports: Coaches measure angles for perfect throws, kicks, and jumps!

Key Takeaways

Sine Rule uses ratios: $a/\sin A$
 $= b/\sin B = c/\sin C$

Cosine Rule uses squares: c^2
 $= a^2 + b^2 - 2ab \cos C$

Both rules solve any
triangle, not just right-
angled!



Keep Exploring Trigonometry!

Trigonometry is a fun way to explore shapes and solve puzzles! Keep practicing and stay curious!



Thank You! Any
Questions?

